Fastener Torques for Reciprocating Compressors

This document lists fastener lubrication and torque requirements for proper assembly of current production Ariel reciprocating compressors (JG:M:P:N:Q:A:R:J:H:E:K:T:C:D:F, KBH:E:K:T:C:D:F:B:V:Z:U, and KB100). For models or components not referenced in this document, see the appropriate Maintenance and Repair Manual. For older units with lower specified torque values, do not assume higher values without consulting the packager and/or Ariel. See manual for specific torque information and detailed assembly procedures.

- Use only Ariel specified fasteners tightened to the correct torque.
- Connecting rod, valve cap, and suction/discharge nozzle (Ariel supplied flange) fasteners are modified to prevent fatigue; do not replace them with standard cap screws. To replace other fasteners with standard cap screws, contact your packager or Ariel.
- Clean and de-burr all threads.
- Do not use molybdenum disulfide lubricants.
- Do not use Never-Seez for fastener lubrication unless specified or excessive stresses may result with specified torques. See Table: Fastener Thread and Seating Surface Lubrication.
- Use anti-seize lubricants sparingly; excessive amounts cause oil analysis to indicate contamination and may unnecessarily increase maintenance costs.
- Re-preserve any fasteners subject to corrosion after installation.

Fastener Thread & Seating Surface Lubrication

NOTE: Lubricate all fasteners both under the head and on the threads.	
Application	**Lubricant**
Forged Steel Cylinders: valve cap, cylinder mounting flange to cylinder body, packing, crank end and head end head, gas passage, Taper-Lok flange	Never-Seez Regular Grade
Cast Cylinders: valve cap, tandem cylinder mounting, packing, crank end and head end head	Mineral oil (ISO 100-150)
Frame: frame to spacer bar not sealed with O-ring or stat-o-seal, KBB:V:Z:U main bearing cap, jack screws. **Pistons:** piston nuts, set screws, piston collar, crosshead threads, tandem piston to piston rod. **All stainless steel fasteners except valve fasteners listed in Table: Hoerbiger Valve Assembly Fastener Torques**	Never-Seez Regular Grade
Connecting rod cap	Lubriplate 630-AA
Crosshead balance nut, crosshead guide to frame to spacer bar, frame to spacer bar sealed with O-rings or stat-o-seals, crosshead guide to frame to main bearing cap, crosshead guide support stud (both ends) and cap nut	**Under the head or mating surface:** Mineral oil (ISO 100-150). **On the threads:** Never-Seez Regular Grade.
All pipe threads	Loctite 567
Fasteners specified with Loctite on threads	Loctite only with no additional lube
All other fasteners, including crankshaft attachments, all O-rings and stat-o-seals	Mineral oil (ISO 100-150)

Bolt Head Grade and Material Identification



Hex Head Grade 5 | Hex Head Grade 8 | Hex Head Grade 9 | Hex Socket Head Grade 8 | 12 Point Grade 8 | 12 Point Grade 5 | 12 Point Stainless 17-4PH | 12 Point Metric Class 8.8

CAUTION:

To replace a fastener, see parts list for fastener grade and part number. Do not use a lesser or greater material grade. Use Ariel parts to replace special fasteners and fasteners with reduced body diameter for

fatigue resistance. Do not torque fasteners with compressor running or pressurized. Read technical manual safety warnings.

Recommendations for Torque Accuracy

1. Critical bolted joints require accurate fastener pre-stress to ensure safe and reliable operation. On Ariel compressors, these joints have specified torque or other engineered tightening methods. Only mechanics properly trained in bolted flange joint assembly should maintain these joints.
2. When torque is specified for a fastener, use a properly calibrated torque wrench to torque the fasteners correctly.
3. Use the correct lubricant on the fastener (see Table: Fastener Thread and Seating Surface Lubrication).
4. Always turn fasteners in by hand until they are finger tight and drawn down evenly. Using a wrench, turn fasteners to draw o-rings into their bores. Using pneumatic, electric, or hydraulic drivers and runners can result in uneven tightening and over-stressed fasteners. If there is significant resistance, remove the fastener to determine and correct the cause.
5. Always apply a steady, slow force to a torque wrench, and stop immediately when the wrench clicks. Do not bounce, jerk, or double tap the wrench – it significantly increases the applied torque. Perform final tightening with a torque wrench.
6. Do not break fasteners loose with a torque wrench - it may overload the wrench and affect its accuracy.
7. Determine torque wrench accuracy range before using - most are not accurate at the extreme lower and higher ends of their range.
8. Clean and de-burr all threads before assembly. Ensure no oil or debris is in the bottom of the bolt hole when installing bolts. Apply oil only to fasteners, not in the bolt hole.
9. Inspect all joint mating surfaces for scratches, nicks, gouges, burrs, corrosion and flatness. This includes the fastener and the fastener contact area on the bolted part. Address any findings before proceeding.
10. **For main bearing cap and connecting rod cap fasteners:** If the bearing cap is stamped with torque sequence numbers, tighten fasteners in that sequence. If not stamped, follow the Figure: Main Bearing Cap or Connecting Rod Cap Torque Sequence.

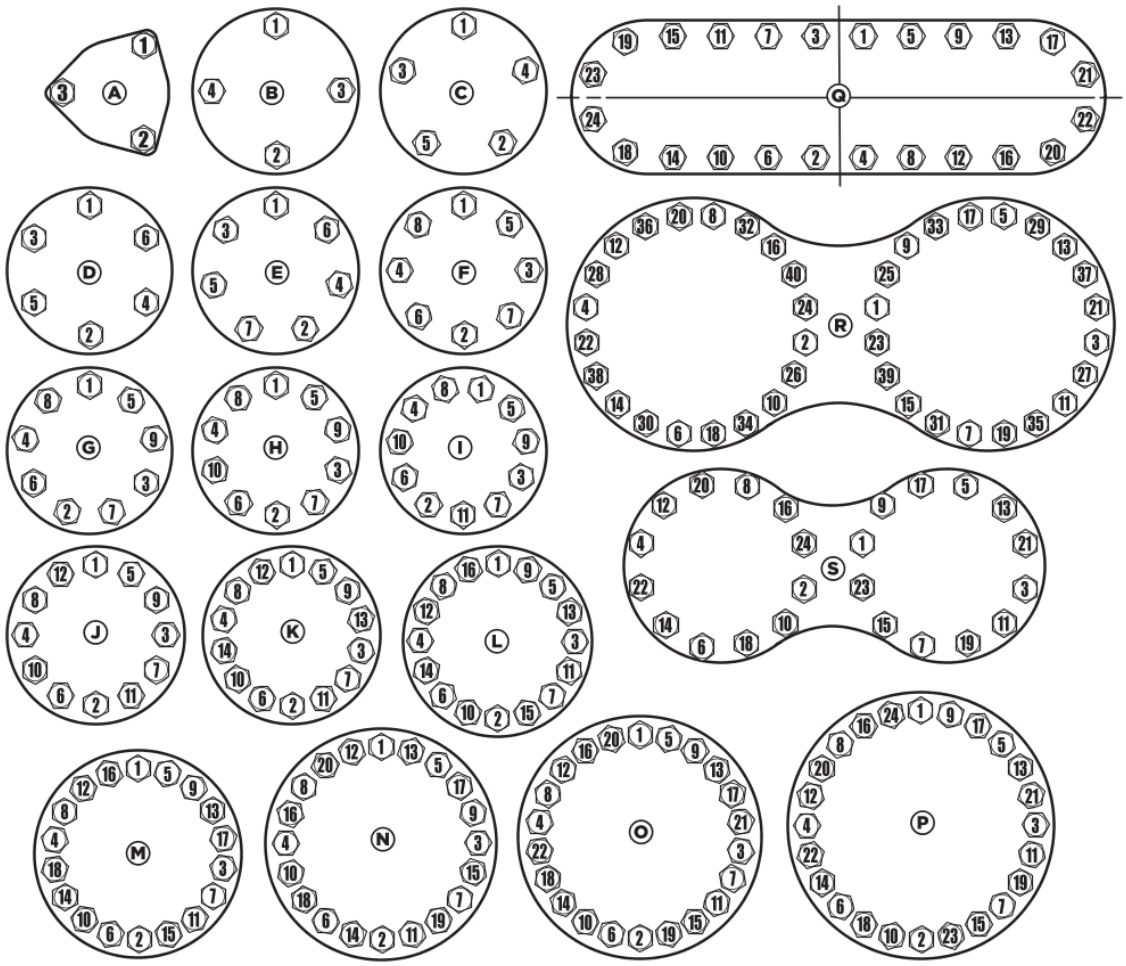
    Main Bearing Cap or Connecting Rod Cap Torque Sequence

    - **Where torque and turn IS NOT SPECIFIED:** Apply 50% of specified torque in the sequence shown to close joint. Once closed, apply 100% of specified torque in the same sequence. Repeat 100% torque in the same sequence until all fasteners stop moving.
    - **Where torque and turn IS SPECIFIED:** Apply 100% of pre-torque in the sequence shown to close joint. Repeat 100% torque in the sequence until all fasteners stop moving. Finally, apply the full turn value to each fastener one time in the same sequence.
11. **For Ariel pressure containing components, see Figure: Pressure Containing Joint Recommended Torque Sequences for recommended torque sequences (includes cylinders, valve caps, cylinder heads, unloaders, gas passage caps, flanges, and packings):**
    - Refer to ASME PCC-1 “Guidelines for Pressure Boundary Bolted Joint Flange Assembly” for commonly accepted practices.
    - Install a new gasket or O-ring.
    - For 3-bolt valve caps, tighten in this sequence: 1-2-3, 2-3-1, 3-1-2. Use the sequence to close the joint without exceeding 25% of full torque. Repeat the three-pass sequence for 25%, 50%, 75%, and 100% of full torque. Repeat 100% torque in a circular pattern until all fasteners stop moving.
    - For other pressure containing joints, tighten the fasteners in four passes using the appropriate sequence for 25%, 50%, 75% and 100% of the specified torque. After 100% torque is reached, continue passes until all fasteners stop moving.

- Inspect the gap under the bolted flange (if visible) after each pass to ensure a consistent gap all around. If the gap is not consistent, the assembly is cocked – disassemble and inspect to determine the cause, and re-assemble after the cause is corrected.

Pressure Containing Joint Recommended Torque Sequences



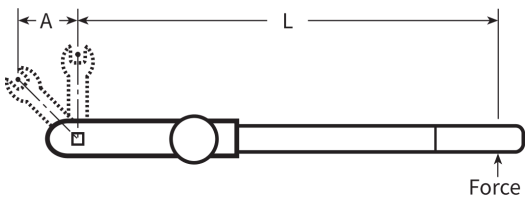
A	3 Fasteners	**E**	7 Fasteners	**I**	11 Fasteners	**M**	18 Fasteners	**Q**	Obround Flange
B	4 Fasteners	**F**	8 Fasteners	**J**	12 Fasteners	**N**	20 Fasteners	**R**	40-Fastener Peanut Flange
C	5 Fasteners	**G**	9 Fasteners	**K**	14 Fasteners	**O**	22 Fasteners	**S**	24-Fastener Peanut Flange
D	6 Fasteners	**H**	10 Fasteners	**L**	16 Fasteners	**P**	24 Fasteners		

12. When finished, reset torque wrench to its lowest setting to relax the spring. A torque wrench left in a high setting stresses the spring and decreases accuracy with time.
13. To determine the torque wrench setting when using a torque multiplier, divide the desired fastener torque by the multiplier actual mechanical advantage, not the design mechanical advantage. See manufacturer documentation for multiplier.
14. If using an offset adapter with a torque wrench, the torque setting is not the actual torque applied, unless the adapter is 90° to the torque wrench. To determine the correct torque setting:

    Tw = Ta x [L ÷ (L + A)]

    Tw = Torque wrench setting, Lb x Ft or N·m.Ta = Torque required at fastener, Lb x Ft or N·m.

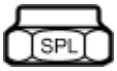
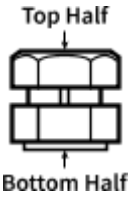
Torque Wrench with Angled Adapter



L = Length of wrench, ft or m (from square drive end to center point of force on handle).

A = Wrench length added by adapter, ft or m (measured to end of adapter on a line parallel to wrench center line).

Hoerbiger Valve Assembly Fastener Torques

Fastener	Nominal Size: Inch - TPI	Type	Torque Lb x Ft (N·m), unless specified
Center Cap Screw - Note a	5/16 - 24	12 Point - Steel Grade 5 Material Parts: SPL3 & 4	14 (19)
	3/8 - 24		24 (33)
	7/16 - 20		35 (48)
	5/16 - 24	12 Point - Steel Grade 5 Material Parts: SPL or SPL5	22 (30)
	3/8 - 24		40 (54)
	7/16 - 20		65 (88)
	5/16 - 24	12 Point - Stainless Steel Grade B8M Material Parts: SPL6	14 (19)
	3/8 - 24		24 (33)
	7/16 - 20		35 (48)
Peripheral Cap Screw	#10 - 32	Hex Socket Head	25 Lb x In (2.8)
	#12 - 28		43 Lb x In (4.9)
	1/4 - 20		110 Lb x In (12)
	5/16 - 18		176 Lb x In (20)
	3/8 - 16		21 (28)
Single-piece Lock Nuts(Greer, Flexloc, Selflock)  (Microlock, Spiralock - Note b)	**Nominal Size**	**Torque Lb x Ft (N·m), unless specified**	
	Inch - TPI	**SPL - Note b**	**Other**
	#12 - 28	96 Lb x In. (11) - Note c	-
	5/16 - 24	15 (20)	109 Lb x In (12)
	3/8 - 24	26 (35)	17 (23)
	7/16 - 20	48 (65)	26 (35)
	1/2 - 20	67 (91)	41 (55)
	5/8 - 18	110 (150)	82 (110)
	3/4 - 16	195 (265)	145 (195)
	7/8 - 14	295 (400)	230 (310)
Center Stud Drake 2-Piece Beam Lock Nut 	**Inch - TPI**	**Bottom Half**	**Top Half**
	1/4 - 28	103 Lb x In (12)	66 Lb x In (7.5)
	5/16 - 24	168 Lb x In (19)	96 Lb x In (11)
	3/8 - 24	16 (22)	96 Lb x In (11)
	1/2 - 20	36 (49) - Note d	20 (27)
	5/8 - 18	73 (99)	40 (54)
	3/4 - 16	130 (175)	70 (95)
	7/8 - 14	210 (285)	115 (155)

a Center cap screw valve assemblies have Spiralock® (SPL) threads to prevent loosening. See bottom of valve assembly for SPL material parts number (3, 4, 5 or 6) and select proper torque from the table. Lubricate both threads and seating surfaces with mineral oil (ISO 100-150) ONLY. If using older valve assemblies not covered in the table, see the original torque chart provided in the compressor tool box, or contact Ariel for instructions. Spiralock® threads cannot be dressed with a standard tap. Clean center cap screws in valve assemblies not marked SPL with Loctite safety solvent and lock them with one or two drops of Loctite #272 thread locking compound. DO NOT use petroleum lubricants.

b Microlock and Spiralock nuts are marked "SPL" on one of the flats.

c Use this torque only with a 17-4PH 900H material center stud.

d Use 29 (39) for 1/2 - 20 bottom half Drake lock nut with non-metallic valve plates in liftwasher valves.

Torques for Specific Applications (Use with Table 4 and Table 5.)

APPLICATION	COLUMN
Dog Point Studs, Seating - Note e	25
• Idler Sprocket Thru Bolt – Prevailing Lock Nut or Hex Lock Nut (Drill & Pin) • Rod Packing - 12 Point Cap Screw or Nut - Note f	40
• Frame to Spacer Bar - 12 Point/Hex Grade 8/9 Cap Screw (except for KBE:K:T:C:D:F:B:V:Z:U Crosshead Guide to Frame to Spacer Bar use Column 55) • KBK:T:C:D:F:Z:U Crosshead Guide Support to Guide - Cap Nut • Sole Plate Pull-up Bolts - Sole plate to guide foot or sole plate to frame foot • Crankshaft Nut Plate/Oil Slinger (JGE:K:T/6, KBB:V) - Hex Cap Screw Grade 9 or 12 Point • FVCP Actuator to Unloader Head - 12 Point Cap Screw, Grade 8 or 17-4PH • FVCP Passage Plug to Unloader Head - 12 Point or Hex Cap Screw, Grade 8/9 or 17-4PH • Roller Thrust Bearing Retainer Clamp Bolt - Ariel Design • Lifting Bracket to Frame - 12 Point Cap Screw	45
Ariel supplied Hex Cap Screw Grade 8/9 or 12 Point or Stud/Nut flange-to-cylinder fasteners: • Valve Cap, Gas Passage Cap, Vent Cap • VVCP, FVCP, or eVCP to Cylinder • Cylinder Head - Note g • VVCP or eVCP Adapter to Head - Note g • Indicator Port Cover • Ariel Supplied Flanged Thermowell • Ariel Supplied Special Flange to Cylinder Including Taper-Lok Flanges • VVCP Seal Carrier • Lube Oil System Components **Note:** For “Peanut” dual nozzle special flanges to cylinder, use Column 60. For other gas nozzle flange types, see instructions from packager.	45
• Crosshead Guide Support to Guide - Hex Cap Screw Grade 8 or 12 Point Cap Screw Grade 8 Head End Cylinder Support to Cylinder – Grade 8 • Distance Piece to Class 11JG Cylinder - Socket Head Cap Screw • Distribution Block Tie Rod - Hex Nut • Rod Catcher to Packing - 12 Point Cap Screw • Tandem Cylinder to Cylinder - Hex Cap Screw Grade 8/9 or 12 Point or Nut • Valve Cap Seal Keeper - Cap Screw; Valve Cap to Retainer - Cap Screw	50
• Crosshead Guide to Frame/Cylinder/Cylinder Mounting Flange/Working Cover, Crosshead Guide to Drive End Cover • Frame to Cylinder • KBB:V:Z:U Crosshead Guide to Frame to Spacer Bar or to Extension Piece • KBE:K:T:C:D:F Crosshead Guide to Frame to Spacer Bar or to Main Bearing Cap • KBH Drive End Cover to Frame - 12 Point Cap Screw • Cylinder Mounting Flange to Forged Steel Cylinder - 12 Point or Nut • Distribution Block Divider Valve - Socket Head Cap Screw - Note h • Unloader Actuator to Valve Cap - Hex Cap Screw, Grade 8	55
• Crosshead Pin Thru Bolt - Hex Nylon Insert Lock Nut. Note: For 5/8-18 Drake Locknut use bottom half: 125 (170), top half: 40 (54). • Auxiliary End Flywheel/Damper/Sprocket Adapter to Crankshaft or spacer to sprocket adapter - Grade 9 or 12 Point Cap Screw • Auxiliary End Spacer/Sprocket Adapter to Crankshaft - Grade 9 or 12 Point Cap Screw • Auxiliary End Sprocket Adapter to Crankshaft - Grade 9 or 12 Point Cap Screw • Crankshaft Vibration Detuner Donut – 12 Point Cap Screw • Drive End Damper to Coupling Adapter - 12 Point Cap Screw	60
• Distance Piece to Crosshead Guide - 12 Point Cap Screw • External-Thrust Bearing Adapter to Crankshaft - 12 Point Cap Screw	65
Piston Rod Oil Slinger - Locknut	70
Piston Rod Oil Slinger - Cap Screw - Note s	80

e In valve cap assemblies with a spring-energized seal, install longer studs with loctite 242.

f Repeat the final torque for rod packing bolts until they no longer turn before the torque wrench clicks.

g Excludes socket head cap screws, which are installed hand wrench tight.

h Step-torque first to 40 Lb x In. (4.5 N•m), and then to 75 Lb x In. (8.5 N•m).

Fastener Torques by Size, U.S. Units - (Shaded Blocks = Lb x In; All Other Blocks = Lb x Ft)

Size: Inch-TPI	COL.25	COL.30	COL.35	COL.40	COL.45	COL.50	COL.55	COL.60	COL.65	COL.70	COL.80
#10 - 24	12	14	17	19	21	24	26	28	31	33	38
#10 - 32	14	17	20	22	25	28	31	34	36	39	45
1/4 - 20	29	34	40	46	52	57	63	69	74	80	92
1/4 - 28	34	41	48	54	61	68	75	82	88	95	109
5/16 - 18	60	72	84	96	108	120	132	144	156	168	192
5/16 - 24	68	82	96	109	123	136	150	164	177	192	18
3/8 - 16	107	128	150	171	193	18	20	21	23	25	29
3/8 - 24	125	150	175	17	19	21	23	25	27	29	33
7/16 - 14	172	17	20	23	26	29	31	34	37	40	46
7/16 - 20	197	20	23	26	30	33	36	39	43	46	53
1/2 - 13	22	26	31	35	40	44	48	53	57	62	70
1/2 - 20	26	31	36	41	46	51	56	61	66	71	82
9/16 - 12	32	38	45	51	57	64	70	76	83	89	100
9/16 - 18	36	44	51	58	66	73	80	87	95	100	115
5/8 - 11	44	53	62	70	Note k	88	97	105	115	125	140
5/8 - 18	51	62	72	82	92	105	115	125	135	145	165
3/4 - 10	79	94	110	125	Note m	160	175	190	205	220	250
3/4 - 16	90	110	125	145	160	180	200	215	235	250	290
7/8 - 9	130	150	180	205	230	255	280	305	330	355	410
7/8 - 12	140	165	195	220	250	280	305	330	360	390	445
7/8 - 14	145	170	200	230	260	290	315	345	375	400	460
1” - 8	200	230	270	305	345	380	420	460	500	535	610
1” - 12	200	255	300	340	385	425	470	510	555	600	680
1” - 14	200	265	310	350	395	440	485	530	570	615	705
1-1/8 - 7	200	325	380	430	485	540	595	650	700	755	865
1-1/8 - 12	200	375	435	500	560	620	685	740	810	870	995
1-1/4 - 7	200	460	540	615	690	770	845	920	1000	1070	1230
1-1/4 - 12	200	520	610	700	780	870	955	1040	1130	1220	1390
1-3/8 - 6	200	600	700	800	900	1000	1100	1200	1300	1400	1600
1-3/8 - 8	200	650	760	870	980	1090	1200	1300	1410	1520	1740
1-3/8 - 12	200	705	820	940	1060	1170	1290	1410	1530	1640	1880
1-1/2 - 6	200	800	935	1070	1200	1340	1470	1610	1740	1870	2140
1-1/2 - 12	200	925	1080	1230	1390	1540	1700	1850	2010	2160	2470
1-5/8 - 12	200	1190	1390	1590	1780	1980	2180	2380	2580	2780	3170
1-3/4 - 5	200	1260	1470	1680	1900	2110	2320	2530	2740	2950	3370
1-3/4 - 8	200	1410	1650	1880	2120	2360	2590	2830	3060	3300	3770
1-3/4 - 12	200	1500	1750	2000	2250	2500	2750	3000	3250	3500	4000
2 - 12	200	2270	2650	3030	3400	3780	4160	4540	4920	5300	6050
2-1/4 - 8	200	3120	3650	4170	4690	5210	5730	6250	6770	7290	8330

U.S. Units

k See ER-112 for 3-bolt valve cap fasteners on 20-1/8, 22, 24-1/8, & 26-1/2 CDZU class cylinders. For 15-7/8Z:U class cylinder crank end head ONLY, use 105 Lb x Ft. All others use 79 Lb x Ft.

m See ER-112.2 for 3-bolt valve cap fasteners on 15-3/8 and 15-7/8 CDFZU class cylinders. All others use 140 Lb x Ft.

Fastener Torques by Size, Metric Units - N•m

Size: Inch-TPI	COL.25	COL.30	COL.35	COL.40	COL.45	COL.50	COL.55	COL.60	COL.65	COL.70	COL.80
#10 - 24	1.4	1.6	1.9	2.1	2.4	2.7	2.9	3.2	3.5	3.7	4.3
#10 - 32	1.6	1.9	2.3	2.5	2.8	3.2	3.5	3.8	4.1	4.4	5.1
1/4 - 20	3.3	3.8	4.5	5.2	5.9	6.4	7.1	7.8	8.4	9.0	10
1/4 - 28	3.8	4.6	5.4	6.1	6.9	7.7	8.5	9.3	9.9	11	12
5/16 - 18	6.8	8.1	9.5	11	12	14	15	16	18	19	22
5/16 - 24	7.7	9.3	11	12	14	15	17	19	20	22	24
3/8 - 16	12	15	17	19	22	24	27	28	31	34	39
3/8 - 24	14	17	20	23	25	28	31	34	37	39	45
7/16 - 14	19	23	27	31	35	39	42	47	50	54	62
7/16 - 20	22	27	31	35	40	45	49	54	58	62	71
1/2 - 13	30	35	42	48	54	60	65	71	77	83	95
1/2 - 20	35	42	49	55	62	69	76	83	90	97	110
9/16 - 12	43	52	60	69	77	86	95	105	110	120	140
9/16 - 18	49	60	69	79	90	99	108	120	130	140	155
5/8 - 11	60	71	83	95	Note n	120	130	140	155	170	190
5/8 - 18	69	83	97	110	125	140	155	170	180	195	225
3/4 - 10	105	130	150	170	Note p	215	235	255	275	300	340
3/4 - 16	120	150	170	195	215	245	270	295	315	340	390
7/8 - 9	175	205	245	275	310	345	380	415	450	480	550
7/8 - 12	190	225	265	300	340	380	415	450	490	525	600
7/8 - 14	195	235	270	310	350	390	425	465	505	545	620
1” - 8	270	310	360	415	465	515	570	620	675	725	825
1” - 12	270	345	405	460	520	580	635	695	750	810	925
1” - 14	270	355	415	475	535	595	660	715	775	830	955
1-1/8 - 7	270	440	515	585	660	730	805	880	950	1030	1170
1-1/8 - 12	270	505	590	675	760	845	930	1010	1100	1180	1350
1-1/4 - 7	270	620	730	830	935	1040	1150	1250	1360	1450	1670
1-1/4 - 12	270	705	825	950	1060	1180	1290	1410	1530	1650	1880
1-3/8 - 6	270	810	950	1090	1220	1360	1500	1630	1770	1900	2170
1-3/8 - 8	270	880	1030	1180	1330	1470	1630	1770	1910	2060	2360
1-3/8 - 12	270	955	1110	1270	1430	1590	1750	1910	2070	2230	2550
1-1/2 - 6	270	1090	1270	1450	1630	1810	2000	2180	2360	2540	2900
1-1/2 - 12	270	1260	1460	1670	1880	2090	2300	2510	2720	2930	3350
1-5/8 - 12	270	1610	1880	2150	2420	2690	2960	3230	3490	3760	4300
1-3/4 - 5	270	1710	2000	2280	2570	2850	3140	3430	3710	4000	4570
1-3/4 - 8	270	1910	2240	2550	2870	3190	3510	3830	4150	4470	5110
1-3/4 - 12	270	2030	2370	2710	3050	3390	3720	4060	4400	4740	5420
2 - 12	270	3080	3590	4100	4620	5130	5640	6150	6670	7180	8200
2-1/4 - 8	270	4240	4940	5650	6350	7060	7760	8470	9180	9880	11300

Metric Units

n See ER-112 for 3-bolt valve cap fasteners on 20-1/8, 22, 24-1/8, & 26-1/2 CDZU class cylinders. For 15-7/8Z:U class cylinder crank end head ONLY, use 140 N•m. All others use 105 N•m.

p See ER-112.2 for 3-bolt valve cap fasteners on 15-3/8 and 15-7/8 CDFZU class cylinders. All others use 190 N•m.

Straight Thread SAE O-Ring Connections into Steel, Cast Iron, or Aluminum: Positionable Lock Nuts, Plugs, & Non-Positionable Fitting Torques

Nominal Fitting Size Inch	SAE Number Size	Thread Size Inch – TPI	Wrench Size Inch	Torque Lb x Ft (N·m)
1/4	-4	7/16 – 20	9/16	160 Lb x In (18)
3/8	-6	9/16 – 18	11/16	22 (30)
1/2	-8	3/4 – 16	7/8	37 (50)
5/8	-10	7/8 – 14	1"	44 (60)
3/4	-12	1-1/16 - 12	1-1/4	70 (95)
1"	-16	1-5/16 – 12	1-1/2	110 (150)
1-1/4	-20	1-5/8 – 12	1-7/8	150 (205)
1-1/2	-24	1-7/8 – 12	2-1/8	155 (210)
2	-32	2-1/2 – 12	2-3/4	220 (300)

Assembled Component Bolt Torques

Compressor Model	Nom. Size inch-TPI	Torque Lb x Ft (N•m)
Main Bearing Cap - 12-Point Cap Screw or Nut		
JG:A:M:P:N:Q, KB100	1/2 - 13	58 (79)
JGR:J	5/8 - 11	97 (130)
JGH:E:K:T, KBH:E:K:T	7/8 - 9	280 (380)
JGC:D:F, KBC:D:F	1-1/8 - 7	540 (730)
KBB:V:Z:U	1-3/8 - 6	350 (475) + 60° - Note q, r
Connecting Rod Cap - Modified Cap Screw		
JGM:P, KB100 - Note w	1/2 - 13	58 (79)
	1/2 - 20	67 (91)
JG:A:N:Q	1/2 - 20	67 (91)
JGR	5/8 - 18	123 (167)
JGJ	3/4 - 16	215 (295)
JGH	7/8 - 14	80 (108) + 90° - Note r
JGH (steel connecting rod)	1" - 14	90 (120) + 90° - Note r
JGE:K:T	1" - 14	See Ariel ER-124.
KBH:E:K:T	1" - 14	120 (165) + 110°- Note r
JGC:D	1-1/8 - 12	See Ariel ER-131.
JGF, KBC:D:F	1-1/8 - 12	165 (225) + 110° - Note r
KBB:V:Z:U	1-3/8 - 12	350 (475) + 110° - Note q, r

Compressor Model	Nom. Size inch-TPI	Torque Lb x Ft (N•m)
Piston Nut - Ariel Design Nut		
JG:A:M:P:N:Q	7/8 - 12	220 (300) - Note s
KB100	7/8 - 16	
KBB:V	2-7/8 - 8	7500 psig (517 barg) & 200 (270) on spanner - Note s, t
JGR:J	1-1/4 - 12	700 (950) - Note s
JGH:E:K:T, KBH:E	1-5/8 - 12	1590 (2150) - Note s, u
KBK:T	1-3/4 - 12	4500 psig (310 barg) & 200 (270) on spanner - Note s, t
JGC:D:F	2 - 12	3970 (5380) - Note s, v
KBC:D:F	2-1/4 - 8	6500 psig (448 barg) & 200 (270) on spanner - Note s, t
KBZ:U	2-1/4 - 12	8700 psig (600 barg) & 30 (40) on pinion drive - Note s, t
Set Screw	5/16 - 24	164 Lb x In (19)
Main Drive Coupling/Flywheel/Drive Adapter to Crankshaft Flange or Flywheel Hub12 Point or Hex Head Cap Screw		
N/A	1" - 8	460 (620)
	1" - 14	530 (715)
	1-1/8 - 12	740 (1010)
	1-1/4 - 12	1040 (1410)

q Use Ariel bearing cap hydraulic torque tool.

r Use Ariel turn indicator.

s Tighten, loosen, then re-tighten.

t Use Ariel piston rod tensioning tool A-23554 for KBB:V, A-19040 for KBZ:U, or A-28362 for KBK:T:C:D:F.

u Or use 3500 psig (241 barg) hydraulic pressure & 50 (68) torque on pinion drive with tensioning tool B-6911. Or use 3500 psig (241 barg) hydraulic pressure & 200 (270) torque on spanner with tensioning tool A-28362 or A-24512.

v Or use 5400 psig (372 barg) hydraulic pressure & 100 (140) torque on pinion drive with tensioning tool B-6911. Or use 5400 psig (372 barg) hydraulic pressure & 200 (270) torque on spanner with tensioning tool A-28362 or A-24512.

w Refer to ER-147 for more information.

Miscellaneous Torques

Fastener		Torque Lb x Ft (N·m), unless specified
ANSI Flange to Cylinder Nozzle		See ER-127.
Compressor to Skid Hold Down Bolts		See packager instructions & Hold-down Bolting to Resist Shaking Forces and Couples in Reciprocating Compressors in Ariel Packager Standards.
JG:A:M:P:N:Q Fan Shaft to Crankshaft		250 (340), 3 times; loosen to hand tight between each tightening.
JGJ Fan Shaft to Crankshaft: 3/8 - 24 TPI		25 (34)
Fenner Drives	P/N A-7352	100 (140)
	P/N A-11560	200 Lb x In (23)
Crosshead Balance Nut	Small Frames, KBH	Slugging Method
	JGH:E:K:T, KBE:K:T, JGC:D:F	3500 psig (241 barg) on Ariel hydraulic tool or slugging method
	KBC:D:F	3500 psig (241 barg) on Ariel hydraulic tool
	KBZ:U	2000 psig (138 barg) on Ariel hydraulic tool
	KBB:V	2800 psig (193 barg) on Ariel hydraulic tool
Filter Cartridge	Top Cover: 3/4 - 10 Hex Nut	75 (100)
	Head: 7/16 - 14 Hex Bolt	40 (54)
Force Feed Lubricator Hex Jam Nut	1" - 14 TPI	75 (100)
	1-1/2 - 12 TPI	112 (152)
Force Feed Lubricator Bearing Housing 1-3/8-20 Left Hand		70 (95) - Use Ariel spanner wrench.
Force Feed Lubricator Pump Mounting1/4-20 TPI Grade 5 Cap Screw		50 Lb x In (5.6)
Auto Relief Valve	1/2 - Ariel Design - Note x	75 Lb x In (8.5)
Auto Relief Valve Manifold Plug	#10 - 24 - Ariel Design	50 Lb x In (5.6)
Grade 5 Hex Cap Screw, Valve Cap Assembly - Cap Nut, Eccentric Vernier Cap 12 Point Cap Screw Grade 8		Up to 40 (54), i.e. hand wrench tight
Rupture Disk Blow-Out Hex Fitting Cap (1/4 Nom. Tube)		40 Lb x In (4.5)
Viscous Damper Filler Pipe Plug (1/8 NPT)		30 (40)
Packing Tie Rod Nut	#10 - 24	20 Lb x In (2.3)
	1/4 - 20 TPI	72 Lb x In (8.1)
	5/16 - 18	144 Lb x In. (16)
KBB:V Strainer Nut: 5/8-11 TPI		70 (95)
Tandem Piston to Piston Rod - Tighten, loosen, then re-tighten to ensure proper torque	5/16 – 24 Stud	18 (24)
	5/16 – 24 Cap Screw	164 Lb x In (19)
	9/16 – 18 Cap Screw	66 (90)
	5/8 – 18 Cap Screw	92 (125)
	3/4 - 16 - Ariel Design	184 (250)
	7/8 - 9 - Ariel Design	230 (310)
Shipping Cover Cap Screw or Nut	1/2 - 13 TPI	22 (30)
	7/8 - 9 TPI	70 (95)
	1" - 8 TPI	120 (165)
	1-1/8 inch or larger	190 (255)
Cylinder Pipe Plug	1/2-inch NPT	Finger tight, then 1-1/4 turns with wrench.
KBH Hoist Ring	1" - 8	130 (176)
KBH Cylinder Cooling Passage Cover	3/8 - 16 Cap Screw	26 (35)

Fastener		Torque Lb x Ft (N·m), unless specified
KBH Tandem Piston Spacer	#10 - 24 Cap Screw	28 Lb x In - Note y
KBH Crosshead Bolt	7/8 - 9 Cap Screw	355 - Refer to KBH Crosshead Torque Sequence below.

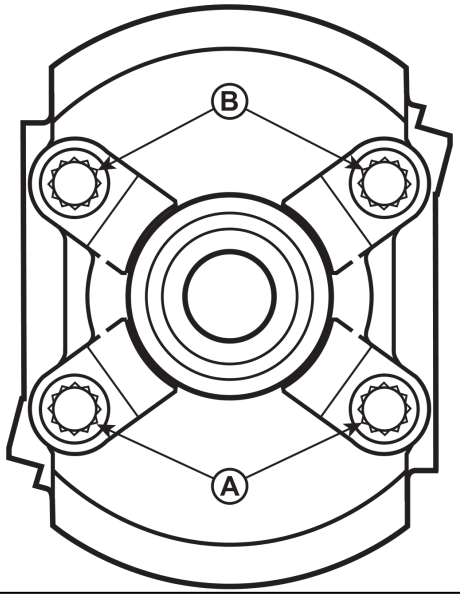
x "1/2" indicates the wrench size. Auto relief valve can swivel in the manifold once torqued. See assembly drawing.

y Stake set screws per manual after tightening.

KBH Crosshead Torque Sequence

1. Torque "A" fasteners to 100 Lbs x Ft.
2. Torque "B" fasteners to 355 Lbs x Ft.
3. Torque "A" fasteners to 355 Lbs x Ft.

KBH Crosshead Fasteners



CHANGE DESCRIPTION: Table Torques for Specific Applications, Column 60: Added 'Drive End Damper to Coupling Adapter - 12 Point Cap Screw'. 202500344: Assembled Component Bolt Torques, Connecting Rod Cap, JGM:P, KB100 - added '1/2-20 67 (91)' and inserted footnote w 'Refer to ER-147 for more information.'